

Enable ISIS functionality on Sonic

Problem Statement

IS-IS (Intermediate System to Intermediate System) is a link-state routing protocol used as an Interior Gateway Protocol (IGP) in TCP/IP networks. It is widely used in large enterprise networks, ISP backbone networks, and data center networks. It calculates the shortest path using the SPF algorithm, supports a hierarchical structure (Level-1/Level-2), and is known for its high scalability, reliability, and fast convergence.

Mentorship Objectives

ISIS, as a widely used IGP protocol, is already supported in FRR, the routing suite used by Sonic. However, Sonic does not yet support directly enabling and using ISIS protocol functionalities. We need to build the initialization, startup, configuration deployment of isisd in Sonic, and implement the ISIS functionality CLI in Sonic.

Expected Outcomes and Deliverables

- Complete the bring-up and initial configuration of isisd on Sonic.
- Finish the Sonic CLI for ISIS functionality.
- Add Sonic-CLI and frrcfgd handling for ISIS configuration and merge changes to SONiC.
- Complete the tests based on 7-nodes vSonic topology.

Recommended Qualifications

- Knowledge of one or more of C/C++, Python.
- Understanding of IP networking and related networking protocols.

Mentor(s) Names and Contact Info

- Songnan Lin, linsongnan.lsn@alibaba-inc.com

Time and location

- Summer - Fall (6 months)
- Location: China

Repo: